

Euclid's Elements — Book I

Proposition 4

Formal Reconstruction — Technical Documentation

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Side-angle-side triangle congruence

1 Introduction

Proposition I.4 is Euclid's first triangle-rigidity theorem. Given two nondegenerate triangles with two corresponding sides and the included angle congruent, it proves the remaining side and the remaining corresponding angles congruent. In modern notation the input is

$$AB \cong DE, \quad AC \cong DF, \quad \angle BAC \cong \angle EDF,$$

and the new conclusions are

$$BC \cong EF, \quad \angle ABC \cong \angle DEF, \quad \angle ACB \cong \angle DFE.$$

Historically, Euclid proves this by superposition. The present Hilbert reconstruction cannot use the physical-language operation of moving one triangle onto the other. At the public Book I level the Lean theorem is only a wrapper around `TriangleCongruentFromSAS`. The audit therefore has to descend below that name.

That descent reveals the main correction to the preserved documentation. The compatibility theorem is itself a wrapper for `SAS`, and the project's full Hilbert Theorem 12 is not merely one primitive SAS axiom call. Hilbert III.5 directly supplies the two remaining angle congruences, but the third side is derived by a separate construction-and-uniqueness argument. The latter constructs a segment witness on a ray, uses SAS once more, invokes the uniqueness clause of angle construction III.4, and finally uses incidence uniqueness to identify the witness with the target vertex.

2 Classical Configuration and Superposition

The classical theorem has no auxiliary construction in the sense of Propositions I.1–I.3. Its complete input consists of two triangles with the SAS data displayed in panel (a) of Figure 1. Euclid then argues by imagining one triangle applied to the other.

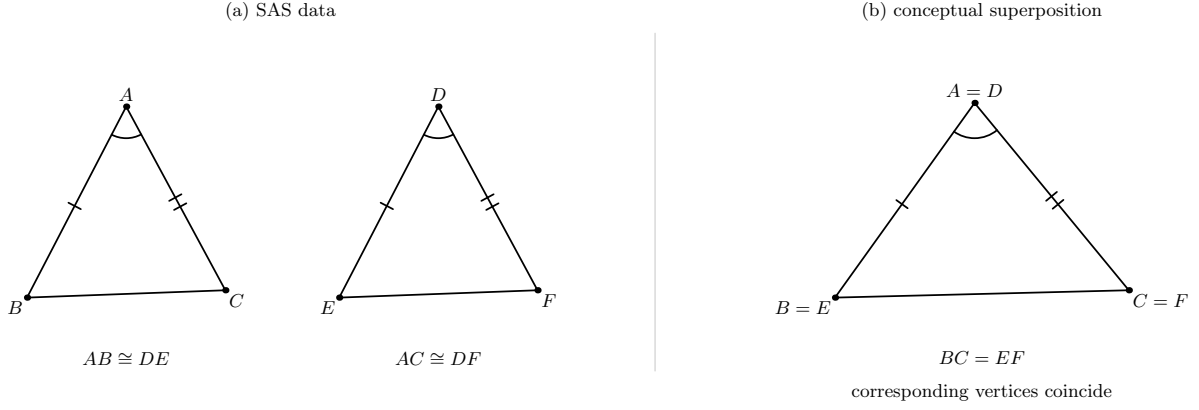


Figure 1: Euclid’s geometric data and the conceptual superposition step. (a) The two triangles satisfy the two side congruences and the included angle congruence. (b) Under Euclid’s superposition, A is placed on D and AB on DE ; the hypotheses force $B = E$ and then $C = F$. The second panel records the conclusion of the conceptual placement, not a Hilbert construction.

This is the only genuinely classical geometric state needed for the theorem, plus the conceptual coincidence state created by superposition. There is no Euclidean point construction to expand into further snapshots.

3 Statement

Theorem 1 (Euclid I.4). *Let ABC and DEF be nondegenerate triangles. Assume*

$$AB \cong DE, \quad AC \cong DF, \quad \angle BAC \cong \angle EDF.$$

Then

$$BC \cong EF, \\ \angle ABC \cong \angle DEF, \quad \angle ACB \cong \angle DFE.$$

Equivalently, all corresponding side and angle data are collected in the project structure $\text{TriangleCongruenceResult}$

The current production declaration is:

```

theorem euclid_proposition_4
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D E F : Geo.Point)
  (hABC : not Collinear Geo A B C)
  (hDEF : not Collinear Geo D E F)
  (hAB : Geo.Congruent A B D E)
  (hAngle : Geo.AngleCongruent B A C E D F)
  (hAC : Geo.Congruent A C D F) :
  TriangleCongruenceResult Geo A B C D E F := by
exact
  TriangleCongruentFromSAS
    Geo A B C D E F
    hABC hDEF hAB hAngle hAC

```

Diagrammatic audit. The preserved chapter used one ordinary triangle picture and one logical flowchart. The triangle picture did not visualize Euclid’s distinctive superposition step, while the flowchart stopped at the compatibility theorem and therefore hid the actual geometric engine below it.

The revised audit replaces those two items by two geometric figures with four panels in total:

$$2 \text{ classical panels} + 2 \text{ formal panels} = 4 \text{ panels in 2 figures.}$$

No panel is introduced for endpoint reversals, noncollinearity permutations, or structure packaging, because those operations do not create a new geometric state.

4 Classical Proof

Euclid begins with the two triangles ABC and DEF . Conceive triangle ABC as placed on triangle DEF so that A falls on D and the side AB lies along DE . Since

$$AB \cong DE,$$

the endpoint B falls on E .

The included-angle hypothesis

$$\angle BAC \cong \angle EDF$$

then makes the ray AC fall along the ray DF . Since

$$AC \cong DF,$$

the endpoint C falls on F .

Thus the third side BC coincides with EF , so

$$BC \cong EF.$$

The remaining angles likewise coincide pairwise, yielding

$$\angle ABC \cong \angle DEF, \quad \angle ACB \cong \angle DFE.$$

The classical proof therefore does not depend directly on Propositions I.1–I.3. Its distinctive proof principle is superposition together with the common notions governing coincidence and equality. Later propositions, starting immediately with I.5, reuse I.4 as a theorem.

5 Formal Proof Architecture

At the public level the chain is short:

$$\begin{array}{c} \text{euclid_proposition_4} \\ \downarrow \\ \text{TriangleCongruentFromSAS} \\ \downarrow \\ \text{SAS.} \end{array}$$

The second arrow matters. The compatibility theorem contains no mathematical argument of its own:

```

theorem TriangleCongruentFromSAS ... := by
  exact SAS Geo A B C D E F

```

The theorem `SAS`, described in the source as Hilbert’s Theorem 12, splits the work into two components:

```

have hAngles :=
  hilbert_sas_remaining_angles
    Geo A B C D E F hABC hDEF hAB hAC hAngleA

have hNeeded :=
  hilbert_sas_third_side_and_angle
    Geo A B C D E F hABC hDEF hAB hAC hAngleA

```

The first component supplies the remaining angles. The second supplies the third side and one of those angles. The final `TriangleCongruenceResult` is only the packaging of these facts.

Thus the deeper dependency is

$$\begin{array}{c}
 AB \cong DE, AC \cong DF, \angle BAC \cong \angle EDF \\
 \downarrow \\
 \text{hilbert_sas_remaining_angles} \\
 \downarrow \\
 \angle ABC \cong \angle DEF, \angle ACB \cong \angle DFE
 \end{array}$$

together with the separate branch

$$\begin{array}{c}
 \text{hilbert_sas_third_side_and_angle} \\
 \downarrow \\
 \text{construct } X \in \text{ray}(EF), EX \cong BC \\
 \downarrow \\
 \text{SAS gives the required angle at } D \\
 \downarrow \\
 \text{III.4 angle uniqueness forces } DX \text{ and } DF \text{ to be one ray} \\
 \downarrow \\
 X, F \in EF \text{ and } X, F \in \text{ray}(D\cdot) \\
 \downarrow \\
 X = F \\
 \downarrow \\
 BC \cong EF.
 \end{array}$$

6 The Hidden Hilbert SAS Engine

6.1 The angle consequences are primitive III.5 consequences

The theorem `hilbert_sas_remaining_angles` applies the primitive congruence axiom `HilbertCongruence.sas` twice. The first application uses the triangles in their stated order and returns the angle at B/E . The second swaps B and C in both triangles and returns the angle at C/F . No new points are constructed in this stage.

This distinction is important: in the present architecture the primitive III.5 mechanism gives the remaining *angles*, not the complete `TriangleCongruenceResult` by itself.

6.2 The third side requires an auxiliary segment witness

To derive

$$BC \cong EF,$$

`hilbert_sas_third_side_and_angle` invokes Hilbert segment construction. Specialized to I.4, it chooses a point X on the ray from E through F such that

$$EX \cong BC.$$

This is the state shown in panel (a) of Figure 2. The original triangle DEF is shown in grey; X is allowed to be a priori different from F . The dashed segment DX is the candidate second ray whose direction will be identified by the subsequent angle argument.

The angle at E between ED and EX is the same angle as the angle between ED and EF , because X and F lie on the same ray from E . The already obtained angle congruence at B/E therefore permits another SAS application, now to the triangles BAC and EDX . It yields

$$\angle BAC \cong \angle EDX.$$

Together with the original hypothesis

$$\angle BAC \cong \angle EDF,$$

the two candidate rays DX and DF make the same prescribed angle with DE on the selected side.

6.3 Angle uniqueness and incidence identify the endpoint

The proof now invokes Hilbert's angle-construction axiom III.4. Internally it constructs a witness Z for the unique ray from D making the prescribed angle with DE . The uniqueness field is applied both to F and to X , showing that

$$\text{ray}(DZ) = \text{ray}(DF) = \text{ray}(DX).$$

But X and F already lie on the same line through E . If $X \neq F$, the line through X, F would be simultaneously the line EF and the line through D and the common ray, forcing D onto EF . This contradicts the nondegeneracy of triangle DEF . Hence

$$X = F.$$

Substituting into $EX \cong BC$ gives the required third-side congruence

$$BC \cong EF.$$

The witness Z has no role in Euclid's classical presentation. It is shown only in the formal panel because it is the carrier of the unique ray returned by the III.4 API. By contrast, no separate figure is created for the two primitive III.5 angle calls: they enrich the original pair of triangles but do not create a new geometric configuration.

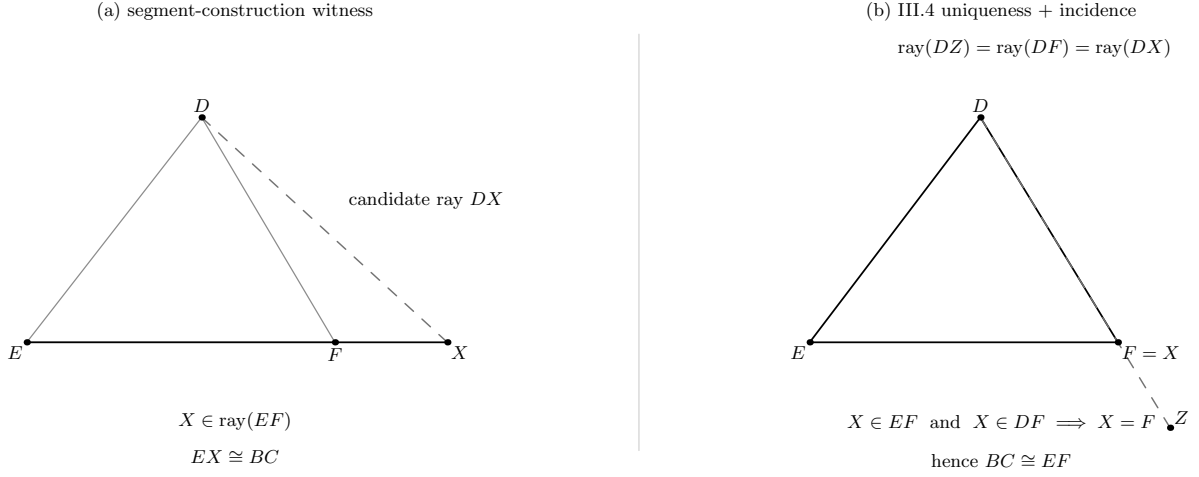


Figure 2: The derived third-side mechanism inside Hilbert Theorem 12. (a) Segment construction places X on the ray EF with $EX \cong BC$. The angle equality later derived for DX is deliberately not drawn while X is still displayed as distinct from F : drawing equal rays already would encode the conclusion. (b) Hilbert III.4 supplies the unique ray represented by DZ ; both F and X lie on it. Together with their collinearity on EF , incidence uniqueness forces $X = F$, hence $BC \cong EF$.

7 Lean Formalization

The production theorem itself is a genuine thin wrapper:

```
exact
  TriangleCongruentFromSAS
    Geo
    A B C
    D E F
    hABC
    hDEF
    hAB
    hAngle
    hAC
```

One level lower, the compatibility declaration is also a wrapper:

```
exact SAS Geo A B C D E F
```

The first substantial theorem is therefore **SAS**. Its source comment explicitly records the architecture: only the third side and the angle at C require the derived construction argument; the remaining fields are hypotheses or direct consequences of III.5.

Inside `hilbert_sas_third_side_and_angle`, the decisive witness is created by

```
rcases HilbertCongruence.segment_construction
  (Geo := Geo) B C E F hBC hEF with
  <X, hRay, hBCX>
```

where the displayed variable names are schematic for the specialization $A' = D$, $B' = E$, $C' = F$. The actual source then uses line incidence, same-side transport, SAS, and `HilbertCongruence.angle_construction` to prove $X = F$.

There is no proposition-local helper theorem, no case split, and no new axiom inside `Proposition04.lean`. The complexity is entirely below the Book I wrapper.

8 Relationship with Earlier Propositions

8.1 No direct dependency on I.1–I.3

The classical proof of I.4 does not invoke Propositions I.1, I.2, or I.3. Those propositions establish early construction machinery, whereas I.4 is a rigidity theorem proved by superposition. They are adjacent in Book I but not direct proposition-level predecessors in the I.4 proof DAG.

The formal theorem likewise does not call the Lean theorems for I.1–I.3. Segment construction used inside Hilbert Theorem 12 belongs to the Hilbert congruence layer, not to Euclid I.2 or I.3 as a Book I dependency.

8.2 Transition to I.5

Proposition I.5 is the first immediate consumer of this rigidity principle. Its classical proof builds auxiliary triangles and applies I.4 twice. In the formal library the same SAS infrastructure is already available below the Book I layer and is reused by the Hilbert theorems that I.5 calls.

Thus I.4 is a structural turning point:

$$\text{I.1–I.3: constructions} \quad \longrightarrow \quad \text{I.4: triangle rigidity.}$$

9 Hilbert Components Used

9.1 Hilbert III.5: primitive SAS angle rigidity

The primitive method `HilbertCongruence.sas` is applied to obtain corresponding angle congruence under the two-side-and-included-angle hypotheses. A second application after swapping the two non-apex vertices supplies the other remaining angle.

9.2 Hilbert III.1: segment construction

The third side is not assumed to follow automatically from the primitive SAS angle statement. A point X is laid off on the ray EF with $EX \cong BC$. This creates the auxiliary triangle used to force the endpoint.

9.3 Hilbert III.4: uniqueness of angle construction

After the second SAS comparison, DX and DF satisfy the same prescribed angle condition on the same side of DE . The uniqueness clause of III.4 identifies their rays.

9.4 Incidence uniqueness

Once X, F lie both on the line EF and on the common ray from D , the uniqueness of the line through two distinct points rules out $X \neq F$. This incidence step is what converts ray uniqueness into equality of the constructed endpoint.

No Euclidean parallel axiom, numerical length, coordinates, or numerical angle measure enters this chain.

10 Formalization Notes

The principal audit correction is foundational rather than cosmetic. The old chapter correctly observed that the public theorem is short, but it collapsed three distinct levels into one phrase, “Hilbert SAS”:

primitive III.5 $\neq$ derived Hilbert Theorem 12 $\neq$ Book I wrapper.

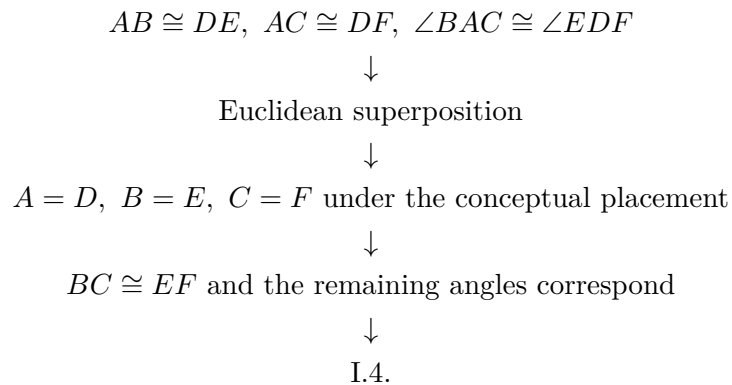
The current code separates them cleanly:

1. `HilbertCongruence.sas` supplies primitive angle rigidity;
2. `hilbert_sas_third_side_and_angle` derives the third side by construction and uniqueness;
3. `SAS` packages the full Hilbert Theorem 12 result;
4. `TriangleCongruentFromSAS` is a compatibility name;
5. `euclid_proposition_4` exposes this infrastructure as Euclid I.4.

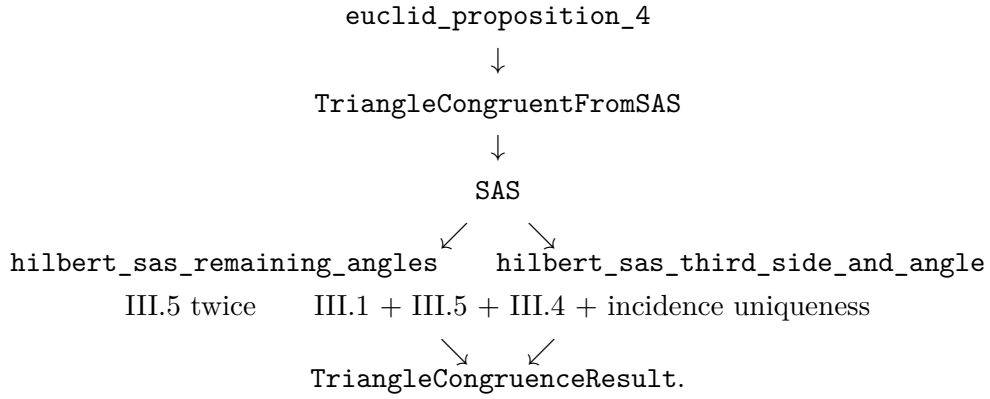
This explains why the Book I proof can consist of a single theorem call without making the underlying geometry trivial. The proof burden has moved below the proposition boundary, but it has not disappeared.

11 Dependency Summary

The classical path is



The formal path is



The audited diagram set mirrors exactly this division: the first figure records Euclid’s superposition logic, while the second records the hidden construction required for the third side in the formal Hilbert theorem.

12 Conclusion

Proposition I.4 is the first major point at which Euclid’s order of exposition and the Hilbert/Lean foundational order diverge. For Euclid, SAS is a theorem proved by superposition. In the reconstruction, primitive SAS angle rigidity belongs to the congruence axioms, while the complete theorem needed by Book I is Hilbert Theorem 12, derived above those axioms.

The current production theorem is therefore short for a precise reason. It is a wrapper around a reusable theorem whose nontrivial third-side proof has already been carried out by segment construction, angle uniqueness, and incidence. The revised figures make both sides visible: Euclid’s conceptual superposition and the actual synthetic construction hidden below the Lean compatibility name.